Air Quality Visual Analytics Individual Report

COMP5048 Assignment 2
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I. Individual Contribution Overview

I was responsible for Task 1, identifying candidate air-quality profiles through coordinated visual exploration without a clustering algorithm.

II. Non Visualisation Contribution

I prepared three view-specific datasets from the team-cleaned source: core_hourly.csv, daily_heatmap.csv and parallel_coordinates.csv. Missing values were not imputed. The multivariate analysis retained 6,941 complete hourly records from 9,357 observations, using timestamp and eight numerical attributes.

I corrected the season labels for Italy and repaired the rebuild-script input and output paths. I also prepared the browser application assets and refreshed the Tableau data sources to support reproducibility.

I completed the initial group-report layout and revised the presentation of Task 1.

III. Visualisation Contribution

I developed three complementary views. The scatterplot shows hourly CO and C6H6 concentrations, coloured by season. The daily heatmap compares scaled medians across eight attributes. Parallel coordinates show hourly profiles on a common 0–1 scale, covering pollutants, sensor responses, temperature and relative humidity.

IV. Interaction and Findings

I configured Tableau highlighting and browser selection controls to explore candidate groups across views. Hourly profiles are matched by timestamp, while the daily heatmap provides context for their corresponding dates. Time and season filters support closer inspection.

The scatterplot reveals a dense lower-left profile and a sparse upper-right profile. The lower profile (CO $\leq$ 1 and C6H6 $\leq$ 5) contains 1,323 hours (19.1%). The upper profile (CO $\geq$ 3 and C6H6 $\geq$ 20) contains 771 hours (11.1%). NOx, NO2 and both sensor responses generally increase in the upper profile. These are visually selected regions within a continuous relationship, not separate algorithmic clusters.

V. Evaluation and Reflection

I evaluated Task 1 by comparing selected profiles across the views, checking the written interpretation against the visual evidence and revising the content following review.

Complete-case selection can exclude periods with missing readings, daily medians hide short events, and overlapping lines make parallel coordinates difficult to read. Environmental conditions may also influence sensor responses.